Network Traffic Analysis Report

RedLine Stealer — Unit 42 PCAP Analysis

Cybersecurity Pre-Master's Portfolio — Week 11

Field	Details
Analyst	Ruben
Date	March 10, 2026
Tool Used	Wireshark
PCAP Source	Unit 42 / Palo Alto Networks — RedLine Stealer Quiz
Malware Family	RedLine Stealer (Infostealer)
Classification	Real-world malware traffic — Educational Analysis

1. Executive Summary

This report documents a network traffic analysis exercise conducted as part of a structured cybersecurity pre-master's portfolio program. The analysis focused on a real-world packet capture provided by Palo Alto Networks Unit 42, containing live network traffic from a RedLine Stealer infection.

RedLine Stealer is one of the most prolific infostealers in active circulation, sold as malware-as-a-service on underground forums. Analysis of the PCAP revealed a complete data exfiltration session: the infected host connected to a command-and-control server over plain HTTP, transmitted detailed system reconnaissance, and exfiltrated browser credentials, cryptocurrency wallet data, API keys, and file contents — all in a single session lasting under 10 seconds.

2. Lab Environment

Component	Details
Analysis Tool	Wireshark (Windows host)
PCAP Source	https://unit42.paloaltonetworks.com/wireshark-quiz-redline-stealer/
Victim IP	10.7.10.47
C2 Server IP	194.26.135.119

C2 Port	12432
Protocol	HTTP (unencrypted) over TCP
Traffic Volume	787 packets / 592KB between victim and C2

3. Traffic Overview

3.1 Protocol Hierarchy

The Protocol Hierarchy (Statistics → Protocol Hierarchy) revealed the following top-level protocols in the capture:

- Ethernet / IPv4 — standard network layer
- UDP / NTP — the infected machine queried time servers, a common malware behavior to accurately timestamp stolen data and verify it is not running in a sandbox
- NetBIOS / SMB — Windows network discovery background traffic
- TCP / HTTP — the primary channel for C2 communication and data exfiltration

The presence of unencrypted HTTP for C2 traffic is characteristic of RedLine Stealer. Unlike many modern malware families that use HTTPS to blend in, RedLine transmits all stolen data in plaintext, making it fully visible in a packet capture.

3.2 Network Conversations

The Conversations view (Statistics → Conversations → IPv4) identified the victim machine at 10.7.10.47 communicating with multiple external IPs. The dominant connection by far was to 194.26.135.119 on port 12432, accounting for 787 packets and 592KB — dwarfing all other connections and immediately identifying this as the C2 server.

Connection	Packets	Data	Significance
10.7.10.47 → 194.26.135.119:12432	787	592 KB	C2 server — primary exfiltration channel
10.7.10.47 → 92.118.151.9	331	373 KB	Secondary external connection
10.7.10.47 → various Microsoft IPs	~20 each	<10 KB each	Normal Windows background traffic

4. C2 Communication Analysis

4.1 Protocol

RedLine Stealer communicates using Microsoft's WCF (Windows Communication Foundation) protocol over HTTP. This is visible in the TCP stream via the tempuri.org namespace references, which is a default WCF service identifier. The malware uses a client-server RPC model with named methods (MSValue1, MSValue2, MSValue3) to structure its data exfiltration.

An authorization token was observed in every request:

```
Authorization: c75a68098b33270de4b87af925fb5ffd
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This token is the victim's unique identifier on the attacker's infrastructure, used to associate stolen data with a specific infected machine.

4.2 System Reconnaissance (MSValue2)

The first major data transfer sent a complete system profile to the C2:

Data Point	Value Observed
Machine ID	3154E784D973FE3C24082CC1E80F37A9
Timezone	(UTC) Coordinated Universal Time
Operating System	Windows 10 Pro for Workstations x64
Username	rwalters
Language	English (United States)
Screen Resolution	1440x900
CPU	Intel Core i5-10200H @ 2.40GHz
RAM	32,765 MB (~32 GB)
Running Process of Interest	mystery_file.ps1 (PowerShell — likely the dropper)

4.3 Targeting Configuration (MSValue2 Response)

The C2 server responded with a targeting configuration instructing the malware what to steal. This is a defining characteristic of RedLine — it is task-driven, receiving instructions from the operator before executing. The configuration included:

- File harvesting: Desktop and Documents — targeting *.txt, *.doc*, *key*, *wallet*, *seed*
- Browser credential theft from 30+ Chromium-based browsers including Chrome, Edge, Brave, Opera, Vivaldi, Yandex, and many others
- Firefox-family browsers: Firefox, Waterfox, Thunderbird, Pale Moon, IceDragon
- Cryptocurrency wallet theft: Metamask, Coinbase, Phantom, Exodus, Electrum, Atomic Wallet, Binance, Coinomi, Monero, Jaxx, and 20+ others

- Browser extension wallets: Yoroi, Tronlink, NiftyWallet, Ronin, Terra Station, and 30+ DeFi extensions
- Environment variable theft: AWS keys, Azure credentials, GitHub tokens, Stripe API keys, Heroku, Docker, Slack, and 50+ service credentials
- Game platform data: Steam, Battle.net, NVIDIA GeForce Experience

4.4 Exfiltrated Data (MSValue3)

The final transmission contained the actual stolen data harvested from the victim machine:

Category	Data Exfiltrated
Browser Credential	rwalters@coolweathercoat.com : My_p@ssw0rd (saved in browser)
Document Exfiltrated	Top_secret_ducment.docx (raw bytes transmitted)
Screenshot	Full desktop screenshot captured and transmitted
Installed AV	Windows Defender (used to assess evasion requirements)
Running Processes	Full process list transmitted including mystery_file.ps1

5. Attack Kill Chain

Phase	Activity	Evidence
Delivery	PowerShell script executed: mystery_file.ps1	Process list in MSValue3 data
Installation	RedLine payload dropped and executed	WCF C2 communication initiated
C2 Establishment	Connection to 194.26.135.119:12432 via HTTP	TCP stream, 787 packets
Reconnaissance	Full system profile sent to C2	MSValue2 — machine ID, OS, CPU, processes
Tasking	C2 sends targeting configuration	MSValue2 response — browser/wallet/file targets
Collection	Credentials, wallets, files harvested	MSValue3 — credentials, document, screenshot
Exfiltration	All data transmitted in plaintext HTTP	TCP stream — raw stolen data visible

6. Indicators of Compromise (IoCs)

Type	Value	Notes
IP Address (C2)	194.26.135.119	RedLine Stealer C2 server
Port	12432	Non-standard HTTP port used for C2
Authorization Token	c75a68098b33270de4b87af925fb5ffd	Victim's unique C2 identifier
Protocol	HTTP WCF / tempuri.org	Characteristic RedLine communication pattern
Dropper	mystery_file.ps1	PowerShell dropper — initial infection vector
Username	rwalters	Compromised account

7. Key Findings and Lessons Learned

7.1 Why RedLine Uses HTTP

RedLine Stealer deliberately uses unencrypted HTTP rather than HTTPS. This is a deliberate trade-off by the malware author — HTTPS would require certificate management and make the malware larger, while HTTP is simpler, faster, and sufficient since the operator controls both endpoints. From a defender's perspective, this makes RedLine significantly easier to detect via network monitoring, which is why SIEM rules and network IDS signatures for RedLine are highly effective.

7.2 The Danger of Infostealer Architecture

The most striking aspect of this analysis is the speed and completeness of the attack. The entire reconnaissance, tasking, and exfiltration cycle completed in seconds. By the time any alert fires, the data is already gone. This highlights why preventative controls (endpoint protection, blocking unknown PowerShell execution) are critical — reactive detection after exfiltration has already occurred is insufficient.

7.3 Connection to Previous Weeks

- Week 7 (Splunk): The C2 IP and port pattern would trigger a Splunk alert for outbound connections to non-standard ports
- Week 8 (Incident Response): The NIST IR process would classify this as a Category 2 (Unauthorized Access) incident requiring immediate credential rotation

- Week 10 (Forensics): The exfiltrated mystery_file.ps1 would be a key forensic artifact — recovering and analyzing it would reveal the full infection chain

7.4 Wireshark Skills Developed

- Protocol Hierarchy analysis to understand traffic composition at a glance
- Conversations view to rapidly identify dominant connections and C2 candidates
- TCP stream following to read complete plaintext C2 conversations
- Display filter usage: http, http.request, ip.addr, tcp.flags.syn
- Live traffic capture and analysis of HTTP traffic from a Windows host

8. Tools and References

Tool / Reference	Purpose
Wireshark	Primary packet capture and analysis platform
Unit 42 RedLine Stealer Quiz	Real-world PCAP source — Palo Alto Networks
MITRE ATT&CK T1071.001	Application Layer Protocol: Web Protocols (C2 via HTTP)
MITRE ATT&CK T1041	Exfiltration Over C2 Channel
MITRE ATT&CK T1555.003	Credentials from Web Browsers